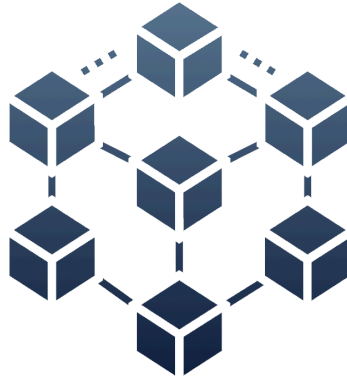




Whitepaper

ECMart: Blockchain-powered Decentralised Non-Fraudulent E-commerce Marketplace

Team Chain Reaction

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Introduction

An online marketplace, often known as an online e-commerce marketplace, is a kind of e-commerce website on which several third parties offer product or service details. Marketplace stakeholders have a variety including investors, employees, customers, and suppliers. With the increasing attention on corporate social responsibility, the concept has been extended to include communities, governments, and trade associations.

Marketplaces have revolutionised people's lives in numerous ways, though it has vulnerabilities that cannot be overlooked. E-commerce marketplaces are exploited with fake reviews, and convince people to buy fake products, which creates trust issues in customers. Using official figures and self-reporting by the world's leading e-commerce sites (including Amazon) on average 4% of all online reviews are fake. Translating this into economic impact, the direct influence of fake online reviews on global online spending is \$152 billion. Fake online reviews influence \$791 billion of e-commerce spending annually in the US, \$6.4 billion in Japan [11]. Besides reported instances of non-payment and non-delivery crimes amounted to \$32.39 billion in 2020 and is projected to reach \$40.62 billion by 2027 - a 25% increase all over the world [8]. Bangladesh also faced challenges in 2021's marketplace - great examples are i) Evaly – it has liabilities over 1000c, 338c taken in advance from the customer, and still owes 206c to merchants, ii) e-orange allegedly embezzled 1100c, also there are dhamaka, Alesha mart, ringID, etc which have committed these crimes [8]. It also had a significant negative impact on Bangladesh's economy.

Avoidance of these multiple notorious incidents has become a necessity in the 21st century as it stunts the economical growth and social stability of a nation. Blockchain technology, which has developed in the last few decades, can be the only optimal solution to address fraud and ensure trust in this landscape. The technology has enabled the creation of smart contracts, which are self-executing contracts stored and executed on blockchain nodes. This allows for the development of secure decentralised e-commerce where the transactions between stakeholder can be secured through tamper-proof escrow contracts. The sensitivity of credit card data theft, data protection, and fraud by centralised marketplaces can be mitigated by establishing e-commerce through blockchain. Moreover, a blockchain-based e-commerce platform can track the supply chain with transparency, eliminate financial fraud, provide non-tamperable review ratings, enable instant transactions and refunds, eliminate centralised controllers, establish trust, and secure financial credentials.

Problem Statement

Despite offering convenience and access to goods and services, the traditional centralised e-commerce system faces significant issues that erode trust, security, and efficiency. These challenges diminish consumer confidence and escalate costs and operational risks for businesses. The key problems identified are as follows:

1. Bangladesh faced significant challenges in its marketplace in 2021, Evaly alone has liabilities of not less than 500 crore taka. Additionally, e-Orange, Alesha Mart, RingID were accused of embezzling around 1100 crore more. [1].
2. Globally, E-commerce fraud remains a critical concern as well, with non-delivery and non-payment offences alone resulting in over \$138 million USD in damages in the USA in 2016 [2].
3. Types of fraud include account takeover fraud, payment fraud, and shipment fraud. In all cases, stakeholders suffer from monetary losses that decline in customer confidence, and reputational harm for e-commerce platforms.
4. Fraudsters can employ credit card theft, and exploit vulnerabilities in online marketplace security due to the marketplace being centralised.

5. Instances of fake or misrepresented products damage trust, while fabricated reviews deceive consumers into purchasing fraudulent goods, intensifying trust issues in the marketplace [3][4].
6. Centralised systems are vulnerable to server failures and cyber attacks, risking business operations and client data. These systems present a single point of failure, making it easier for hackers to access user data [3].
7. Authorities of centralised marketplaces can manipulate information, change prices, and alter terms of sale without transparency.
8. Centralised control can lead to monopolies, restricted competition, and higher consumer costs, exemplified by Amazon charging around 8% and eBay 10% of the final sale price [2].

Solution

In today's rapidly evolving digital landscape, the necessity for a transformative e-commerce solution has never been more pronounced. Centralised platforms wield monopolistic control, issues of sensitive data leakage and jeopardising consumer-supplier trust through financial fraud. Transitioning to a decentralised e-commerce model promises a paradigm shift: one that ensures heightened security, transparency, and efficiency through blockchain technology.

The proposed web3 marketplace can establish a technology-driven business which leverages blockchain on 4 key factors - secure payment escrow on product sell/purchase, tamperproof review/rating, instant monetization, tracking shipment by establishing a secure, transparent, efficient platform for e-commerce transactions, product reputations, and dispute management by eliminating single point-of-failure and facilitating direct peer-to-peer interactions. It features a robust rating and review system that enhances trust and accountability by gathering user feedback. Payments are securely held in escrow until buyers confirm receipt of products, effectively preventing fraud. These advancements foster a peer-to-peer marketplace that empowers users with enhanced security and confidence, revolutionising the traditional e-commerce model. This evolution not only addresses current market inefficiencies but also sets the stage for a more equitable and resilient digital economy, benefiting both consumers and businesses alike.

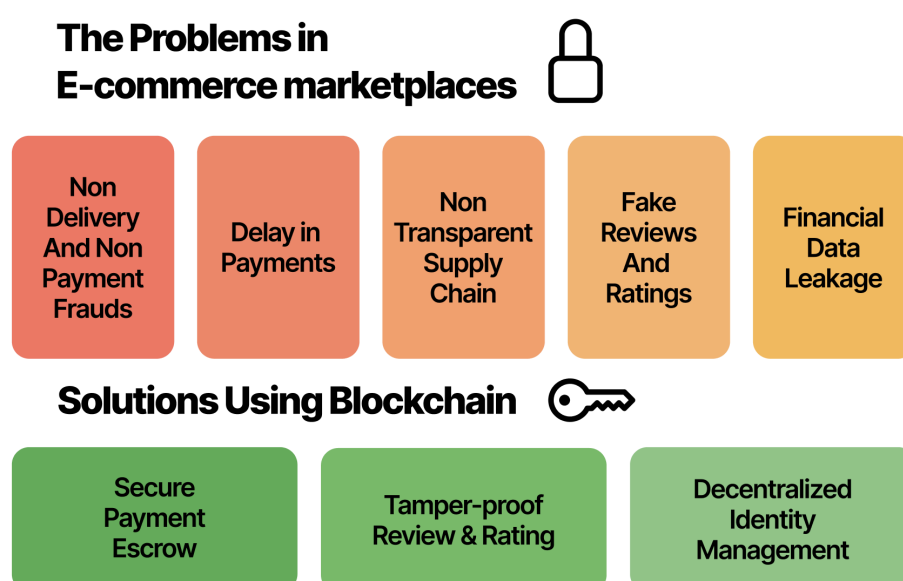


Figure 1: Impact of Blockchain in e-commerce

Why Blockchain in e-commerce?

The global e-commerce, especially Bangladeshi e-commerce (like Evaly), landscape is plagued by fraud and a lack of trust. Centralised platforms create vulnerabilities, hindering growth. Blockchain technology emerges as a secure, transparent, and decentralised solution.

Blockchain can revolutionise e-commerce in diversified ways, such as:

- **Lower transaction costs** benefit both buyers and sellers.
- **Secure escrows** eliminate non-payment and non-delivery fraud.
- **Trackable supply chains** ensure product authenticity.
- **Immutable reviews** build trust and prevent manipulation.
- **Instant transactions** streamline the e-commerce experience.
- **Decentralisation** empowers users and fosters competition.
- **Faster processing** reduces dispute resolution time.
- **Transparency** builds trust between buyers and sellers.
- **Data protection** safeguards user privacy.

In Bangladesh, with its booming mobile-first economy, blockchain presents a unique opportunity to address these challenges and unlock the true potential of e-commerce.

Catalyst for Bangladesh's Socio-economic Transformation

Under the visionary leadership of the current government, Bangladesh is ambitiously striving to elevate its economic status and ultimately attain the recognition of a developed nation. This transformative journey is guided by a strategic framework aimed at fostering economic growth, enhancing technological innovation, and ensuring inclusive development across all sectors of society. Our blockchain-based e-commerce platform aligns perfectly with these aspirations, demonstrating its potential to contribute significantly to Bangladesh's socio-economic transformation. Here's how:

Digital Economy and ICT Development:

Our platform plays a crucial role in advancing the nation's technological landscape by offering secure, transparent, and efficient transactions. The integration of blockchain technology enhances the growth of the ICT sector, contributing significantly to the expansion of tech exports and solidifying Bangladesh's position in the global market.

Economic Growth and Employment:

Our platform can stimulate economic growth by reducing fraudulent activities in e-commerce, thereby increasing consumer and investor confidence. Additionally, it can create new jobs in blockchain development, cybersecurity, and digital marketing.

Poverty Eradication and Financial Inclusion:

By offering a transparent and safe platform for transactions, this technology may support the growth of small enterprises and entrepreneurs, boosting their revenue and assisting in the fight against poverty. Financial inclusion is promoted by blockchain's transparency, which guarantees equitable access to financial services.

SDGs

The Sustainable Development Goals (SDGs) may benefit greatly from the blockchain-based e-commerce platform. The illustration of SDGs and its benefit from our blockchain-based platforms are given below:



Figure 2: Sustainable development goals addressed through ECMart

SDG 8: Decent Work and Economic Growth:

- **Impact:** By decreasing e-commerce fraud and boosting investor and customer confidence, the platform can promote economic growth.
- **Alignment:** Encourages full and productive employment, equitable and sustainable economic growth, and decent work for all.

SDG 9: Industry, Innovation, and Infrastructure:

- **Impact:** The platform encourages innovation and the creation of strong digital infrastructure by utilising blockchain technology.
- **Alignment:** Promotes inclusive and sustainable industrialization, strengthens infrastructure, and stimulates innovation.

SDG 12: Responsible Consumption and Production:

- **Impact:** By improving supply chain transparency, the platform enables customers to make well-informed decisions about the goods they buy.
- **Alignment:** Upholds the integrity and openness of supply networks to encourage sustainable patterns of production and consumption.

SDG 17: Partnerships for the Goals:

- **Impact:** In order to accomplish shared objectives, the platform may promote collaborations between different stakeholders, such as companies, governments, and communities.
- **Alignment:** Fortifies the methods of execution and resurrects the international alliance for sustainable development.

Features

This innovative blockchain-based e-commerce application, built on Scale blockchain network using Hardhat, Solidity, BigchainDB, and IPFS, offers a secure and transparent marketplace

experience through a comprehensive suite of features designed to empower both buyers and sellers. Here's a breakdown of its functionalities:

- **Decentralised Identity Management:** User identities are managed through secure public key cryptography during registration. Personal information is encrypted and stored in a decentralised database, minimising the risk of data breaches.
- **Secure Transactions with Smart Contracts:** Smart contracts facilitate secure transactions with an escrow service. Funds are held until both buyers and sellers fulfil their obligations, minimising fraud.
- **Transparent Review and Rating System:** Only verified buyers can leave reviews and ratings, ensuring authenticity. Reviews are based on factors like delivery speed, communication, product quality, and seller behaviour.
- **Real-time Order Tracking:** Track your product's journey throughout the entire supply chain. Delivery parties authenticate product reception using a verification code, updating the blockchain with the product's current shipping status.
- **Decentralisation and Data Security:** The application leverages cryptography and distributed ledgers to create tamper-proof, secure applications.
- **Enhanced Security:** The application is immune to cyber-attacks like Sybil attacks, DDoS attacks, social engineering, routing attacks, and exchange hacks.
- **Generic E-commerce Features:** Registration, product browsing and listing, ordering, logistic delivery management, and secure communication between buyers, sellers, and delivery providers.

These features foster trust, transparency, and security, creating a robust and user-friendly e-commerce experience.

Market-Size

National Opportunity

The business-to-customer e-commerce market in Bangladesh is expected to grow by 17.61% on an annual basis, and reach Tk65,966 crore in 2022. The e-commerce market size in 2021 was about Tk56,870 crore and will be around Tk1.5 lakh crore by 2026 [5]. Moreover, according to statista, the estimated e-commerce users of Bangladesh is around 11 million who shop from online platforms very often [13]. This surge underscores the resilience and growth potential of the e-commerce landscape in Bangladesh, fueled by changing consumer habits and robust digital infrastructure [9].

International Opportunity

The widespread availability of the internet, along with increased access through smartphones and other devices, has enabled a larger portion of the global population to connect and shop online. Around 2.4 billion people all across the world, which is around 30% of the entire world's population, are currently using e-commerce in their day to day life [12]. In 2023, the market size was valued at USD 16.27 trillion, and it is projected to grow to USD 57.22 trillion by 2032, with a compound annual growth rate (CAGR) of 15% from 2023 to 2032. The Asia-Pacific region is identified as the largest market [10].

As e-commerce transactions involve the exchange of sensitive personal and financial information, the risk of cyberattacks, data breaches, and identity theft is a major concern. According to secondary analysis, Account take-over (ATO), which increased by 131% in the second half of 2022 compared to H1 in 2021, continues to be an e-commerce fraud issue in 2023. Thus, the growing cybersecurity concern is expected to hamper the market growth over the forecast period [10].

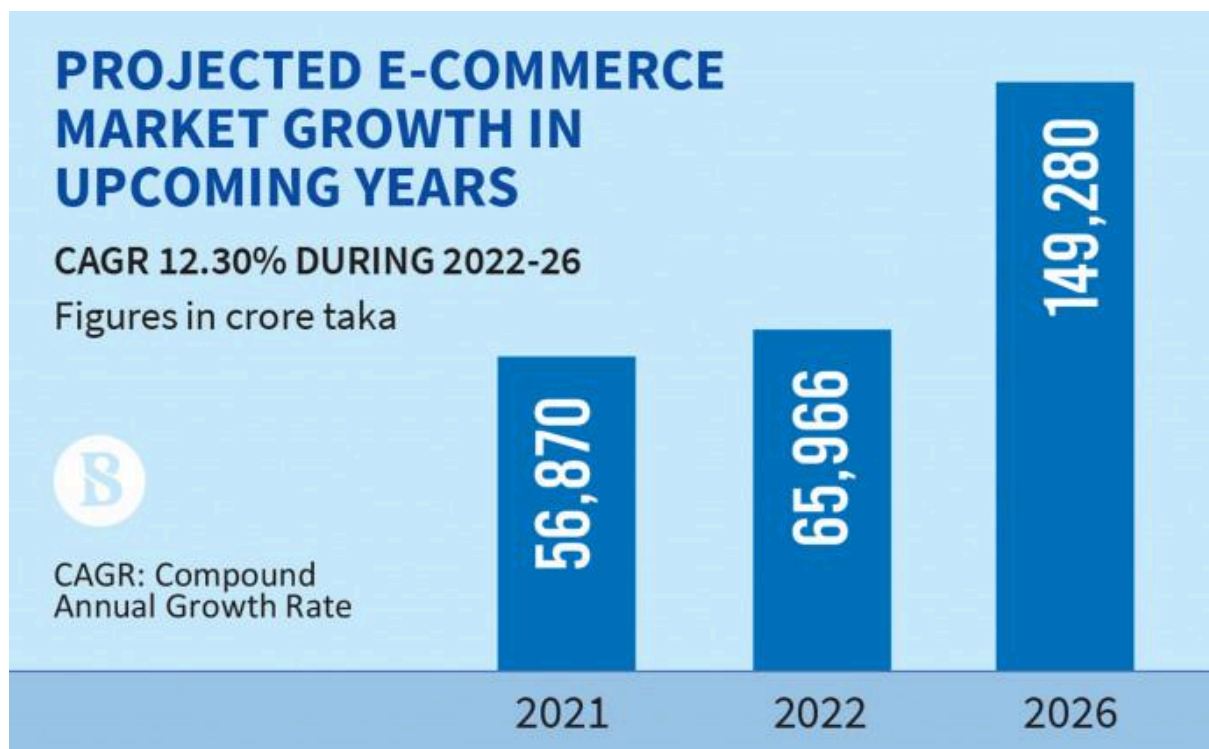


Figure 3: E-commerce Marketplace Sales in Bangladesh, amount in Crore(tk) (Source: The Business Standard [5])

Partners

Partners Working together with important stakeholders is crucial for the effective implementation and management of the blockchain-based e-commerce platform. Possible companions consist of the following.

Local Delivery Services: Effective and dependable delivery services can be guaranteed throughout the nation by partnering with well-known logistics firms like RedX and Pathao. These collaborations will improve customer satisfaction and enable real-time order tracking.

Government Organisations: Collaborating with government bodies such as the *Bangladesh Telecommunication Regulatory Commission (BTRC)*, which regulates telecommunications and internet services, and *Bangladesh Bank*, which oversees online payment systems, ensures secure financial transactions. The *ICT Division* focuses on digital infrastructure and technology, promoting digital entrepreneurship, while the *Ministry of Commerce* creates policies for commerce, including e-commerce. The Bangladesh Bank is a vital partner for this next generation of e-commerce to stand out because the cryptocurrency ban needs to be resolved with proper regulation. Hence, it would fuel both ECMart, and other blockchain-based FinTech applications to expand in the country's horizon.

Private Institutions (e.g., e-CAB): Partnering with private institutions like e-CAB, the e-Commerce Association of Bangladesh, fosters industry collaboration, and facilitates wider adoption within the Bangladeshi e-commerce ecosystem.

Marketing and Influencers: Working with influencers and marketing firms can help to publicise the platform and foster user confidence. To promote acceptance of blockchain technology in e-commerce and educate potential users about its advantages, influencers can be very helpful.

Competition & Risks

Competitive Landscape

While Bangladesh boasts a thriving e-commerce sector, existing platforms like Daraz and international giants like Amazon operate on centralised models. This creates vulnerabilities to fraud, data breaches, and manipulation. Our proposed blockchain-based e-commerce marketplace presents itself as a first-mover in the Bangladeshi context, directly competing with established players but offering a fundamentally more secure and transparent alternative.

Value Proposition

Compared to traditional e-commerce platforms, our solution offers significant advantages:

- **Enhanced Security:** Our blockchain architecture is immune to common attacks like Sybil attacks, DDoS attacks, social engineering, routing attacks, and exchange hacks, fostering a secure environment for both buyers and sellers [6].
- **Decentralised Identity Management:** User identities are managed through secure public key cryptography, eliminating the risk of centralised data breaches. Encrypted personal information stored on a decentralised database provides additional security [7].
- **Transparent Review and Rating System:** Only verified buyers can leave reviews, ensuring authenticity and fostering trust in seller interactions.
- **Secure Transactions with Smart Contracts:** Smart contracts automate transactions with an escrow service, holding funds until both parties fulfil their obligations, minimising fraud and disputes.
- **Real-time Order Tracking:** Buyers can track their products throughout the entire supply chain, enhancing transparency and trust in the delivery process.

Business Risk Mitigation

- **User Adoption:** Building user awareness and trust in a new technology like blockchain will be crucial. We plan to achieve this through targeted marketing campaigns, educational initiatives, and partnerships with established players in the Bangladeshi e-commerce ecosystem.
- **Regulation:** As blockchain technology is still evolving, regulations in Bangladesh might not be fully established. We will closely monitor regulatory developments and ensure our platform adheres to all current and future regulations.
- **Integration with Existing Infrastructure:** Integrating our blockchain platform with existing payment gateways and logistics providers in Bangladesh will be essential for seamless user experience. We will actively seek partnerships with these stakeholders to facilitate smooth integration.

Risk Management

- **Non-Cooperation of Partners:** Offering exclusive benefits or early access to the platform to incentivize participation for their first 90 days can address the issue.
- **Non-Alignment of Incentive:** Designing a tokenized reward system for Buyers, when they earn tokens for verified purchases and reviews, sellers earn tokens for completing orders with high ratings. Tokens can also be used for discounts or platform upgrades.
- **Lack of Business Traction:** Develop a targeted marketing campaign with a strong social media presence. Partner with Bangladeshi influencers (like, *Salman Muqtadir*)

to promote the platform and its benefits. Offer limited-time discounts and referral bonuses to attract early adopters.

- **Other Business Risks:** Continuously monitor market trends and competitor actions. Conduct regular SWOT analyses to identify strengths, weaknesses, opportunities, and threats. Develop contingency plans to address potential disruptions.

Performance Advantage

- **Security:** Our decentralised architecture offers unparalleled security compared to traditional e-commerce platforms, attracting security-conscious users and businesses.
- **Transparency:** The immutable ledger ensures complete transparency in transactions and product information, fostering trust between buyers and sellers.
- **Efficiency:** Smart contracts automate processes, streamlining transactions and reducing administrative overhead for both buyers and sellers.
- **Traceability:** Real-time order tracking on the blockchain provides superior visibility and control over the supply chain, reducing delays and improving efficiency.

Technical Risk Mitigation

- **Scalability:** Scalability is a potential challenge for blockchain platforms. We will leverage Skale's elastic sidechain technology to ensure our platform can accommodate future growth.
- **Interoperability:** Ensuring seamless interoperability with different blockchain protocols might be necessary. We will adopt open-source standards and protocols to facilitate smooth future integrations.
- **Technical Expertise:** Building and maintaining a robust blockchain platform requires specialised expertise. We will assemble a team of skilled blockchain developers to ensure the technical integrity of our platform.

Architecture & System Overview

Tech-Stack

- **Front-End Development:** The user interface utilises a combination of HTML for structure, Tailwind CSS for rapid development, and JavaScript for interactivity.
- **Blockchain Integration:** User interaction with the blockchain is facilitated through the popular Metamask wallet. Ethers.js provides a bridge between the front-end and the Ethereum blockchain for functionalities like data access and manipulation within smart contracts.
- **Data Storage:** IPFS is used as a distributed data storage to store the e-commerce media file.
- **Back-End Development:** BigchainDB, a scalable blockchain database, securely stores marketplace data like order information, user credentials, and product details with corresponding blockchain addresses. Express.js, a lightweight web framework, forms the core of the back-end server, enabling essential e-commerce functionalities through robust APIs. EJS streamlines dynamic content generation by allowing the integration of server-side logic and data with the front-end UI.

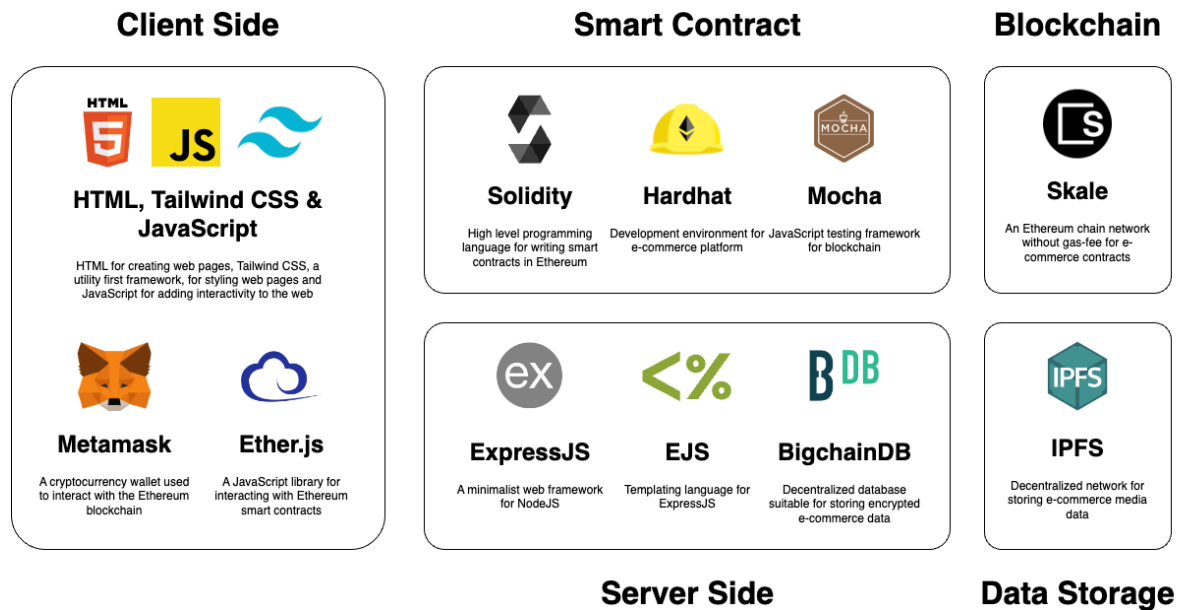


Figure 4: Tech-Stack of Blockchain-based e-commerce

- Smart Contract Development and Testing:** Solidity, a high-level programming language, is used to develop the smart contracts that govern marketplace operations. Hard Hat provides a comprehensive development environment for writing, deploying, and testing these smart contracts on the SKALE blockchain. Finally, the Mocha testing framework ensures the reliability and integrity of smart contracts before deployment.

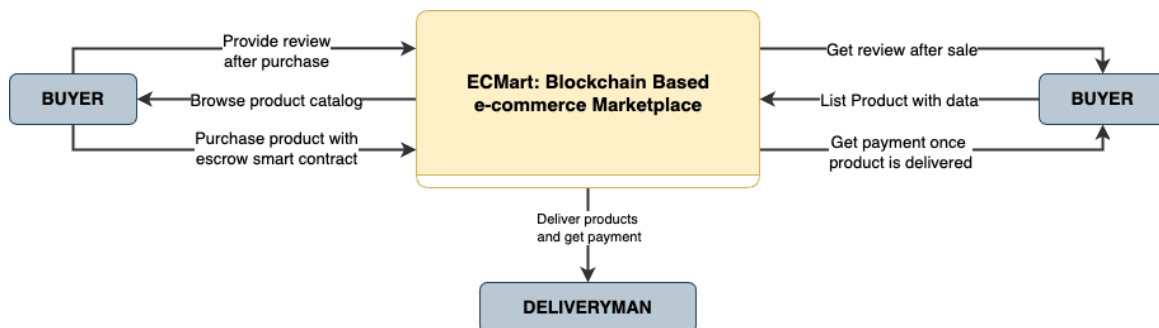


Figure 5: Context Diagram of Blockchain-based e-commerce.

System Overview

The registration of stakeholders (buyers, sellers, delivermen) is done both on off-chain and on-chain. The public address of the user's **Metamask** account is used to register them on the blockchain network. There are a total **3 main smart-contracts** - *Product*, *Order*, *EC Mart*. Every time a seller add a product from his dashboard the product details is stored in decentralised off-chain **BigchainDB** database, also a new product contract is created in the **SKALE blockchain** using the details provided and the new product contract address is stored in the main **EC Mart** to keep it in the blockchain marketplace, the media of the product data is distributed storage **IPFS**. Buyers can view the product from an off-chain **BigchainDB** database using the backend-server made using **Express.js**. Moreover, in case of, off-chain server failure, the product data can be fetched from the blockchain using the **EC Mart** Contract. Buyers can order products using the user interface; and then, similar to product contract, a new order contract is generated in the blockchain once buyer place an order

(consisting with several products) and the order details are also stored in off-chain **BigchainDB** and IPFS.

Buyer, sellers, and delivery men who are associated with the order can view the order placed in the system from both off-chain and on-chain. The delivery men can view the upcoming orders, the first (delivery man) to accept the order using their **Metamask wallet** address will be assigned to deliver the product. A tracking is performed while the delivery man (by cross-checking the product and quantity) receives the product from the seller as he has to perform a transaction via his **Metamask** wallet from his dashboard, and the seller will confirm it during the product handover via his blockchain wallet.

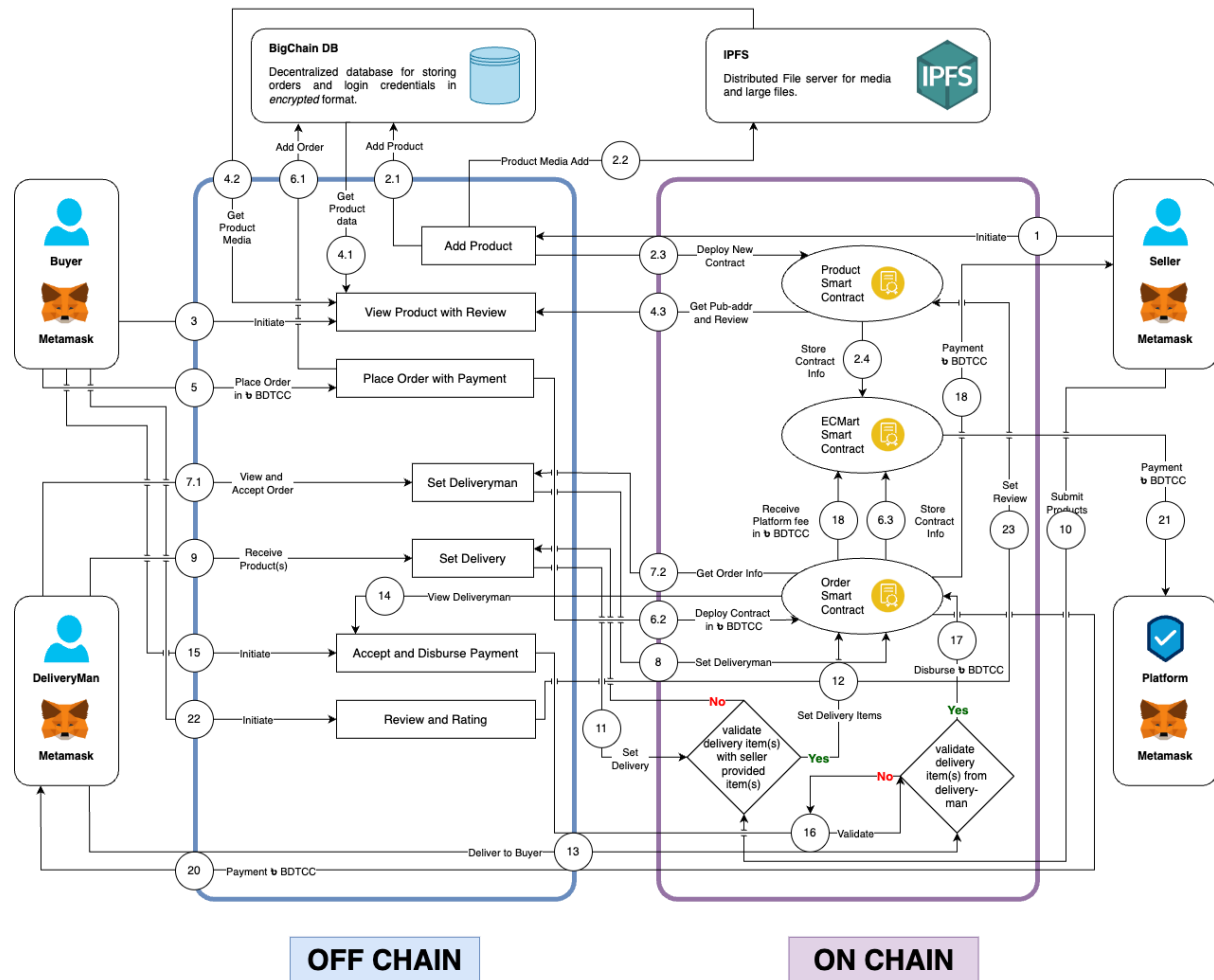


Figure 6: System Diagram of Blockchain e-commerce Marketplace

Buyer, sellers, and delivery men who are associated with the order can view the order placed in the system from both off-chain and on-chain. The delivery men can view the upcoming orders, the first (delivery man) to accept the order using their **Metamask wallet** address will be assigned to deliver the product. A tracking is performed while the delivery man (by cross-checking the product and quantity) receives the product from the seller as he has to perform a transaction via his **Metamask** wallet from his dashboard, and the seller will confirm it during the product handover via his blockchain wallet.

In the similar fashion, Buyer will verify the product received, checking the quantity and product received from the delivery man. In case of wrong/no product with wrong/no quantity buyer can inform the delivery man to initiate correction using 'Set Delivery' method on his

dashboard. Upon confirmation by the buyer, he has to trigger fund disbursement from his dashboard, only then the delivery man can have the status of the product delivered. The order status updated both on **BigchainDB**, and Order **smart-contract** of **SKALE** chain. During the payment disbursement, the sellers of the order, and the delivery man will receive the payment, and ECMart will receive its platform incentive.

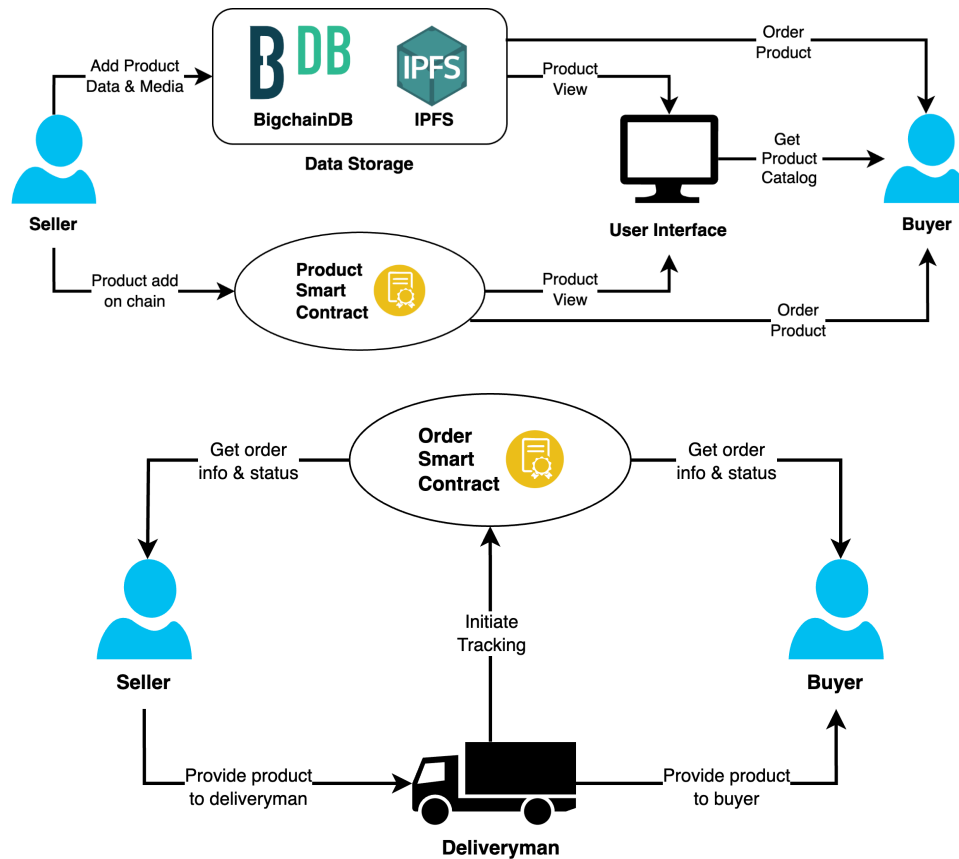


Figure 7: Process of Listing, Viewing, and Ordering in the Blockchain-based e-commerce

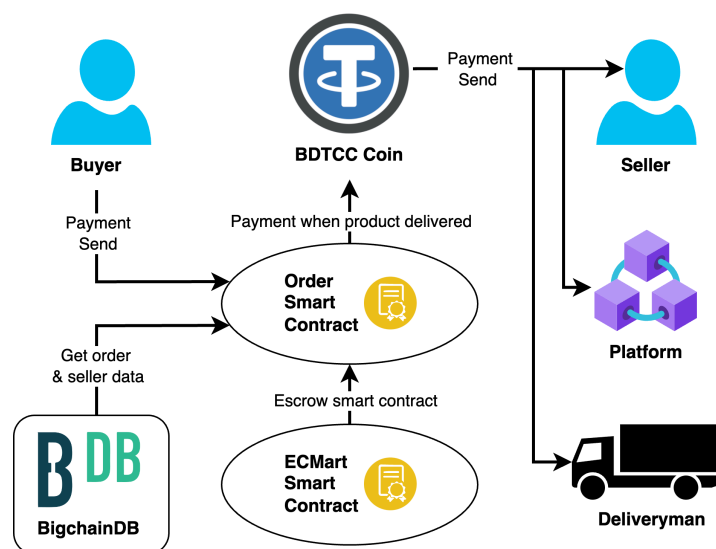


Figure 8: Process of Payment, Fund Disbursement, and Delivery in the Blockchain-based e-commerce system

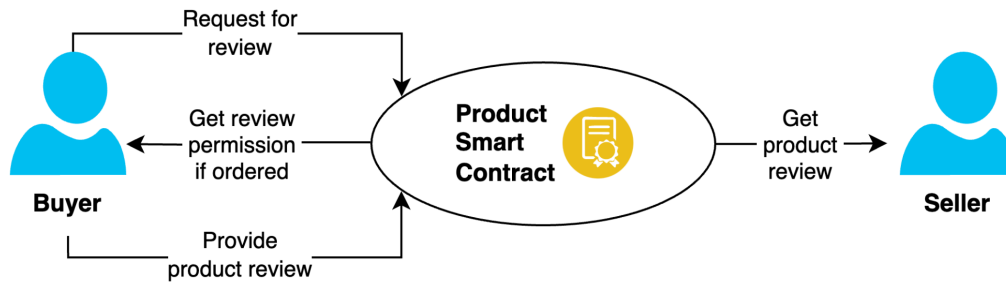


Figure 9: Process Review in the Blockchain-based e-commerce system

Finally, the buyer can provide review-rating on his products from the frontend, hence, the review and rating is send on the blockchain of its corresponding product contracts by verifying the authenticity of the purchase using order **contract** as the buyer need to send the review-rating through his same **Metamask wallet** he used during the purchase of the product. Any platform users can view the review-rating secured by blockchain while product browsing.

Implementation

The prototype implementation of the proposed e-commerce blockchain marketplace is detailed in **Appendix A**.

Architecture

The above figure shows the architecture of smart contracts associated with the proposed blockchain-based e-commerce marketplace. **Diamond proxy pattern**, for building highly modular and upgradeable smart contracts by separating functionalities into smaller, independent facets, is used for handling the heavy methods and variables of the e-commerce marketplace. The diamond contract receives all the transaction calls sent to the marketplace contracts, then it **delegates the call** into corresponding facets which then map the corresponding contract to complete the blockchain transaction, or provide the information required from the contract.

Why choose SKALE Blockchain?

While a public blockchain like Ethereum offers advantages like decentralisation and security which is necessary for an open marketplace to operate, its high transaction fees can hinder e-commerce viability. Therefore, we utilise the SKALE Network, a public blockchain built on top of Ethereum. SKALE offers the security and transparency of Ethereum while providing significantly lower transaction fees, making it ideal for our high-volume e-commerce platform.

Interaction with Non-Blockchain Systems or Legacy Systems

The front-end user interface will be built using traditional web technologies (HTML, CSS, JavaScript) and interact with the back-end server. The back-end server, potentially built with Node.js and Express.js, will handle communication with the blockchain and other external systems. Metamask, which is the only content that needs to be supported to make the application work in legacy systems, is supported by legacy versions of Chrome, Firefox, Brave browsers.

How is the transaction verified?

In the Skale network, every ecommerce transaction is first submitted by 'Proposal Module' for consensus, which is a hybrid consensus of PoW(Proof of Work) and PoS(Proof of Stake),

then it is validated by 'Validation Module'. Finally, it is broadcasted to all the nodes of the network using the 'Broadcast Module'. However, the consensus protocol & validation mechanism of SKALE is necessary for making the platform 'Gas' charge free.

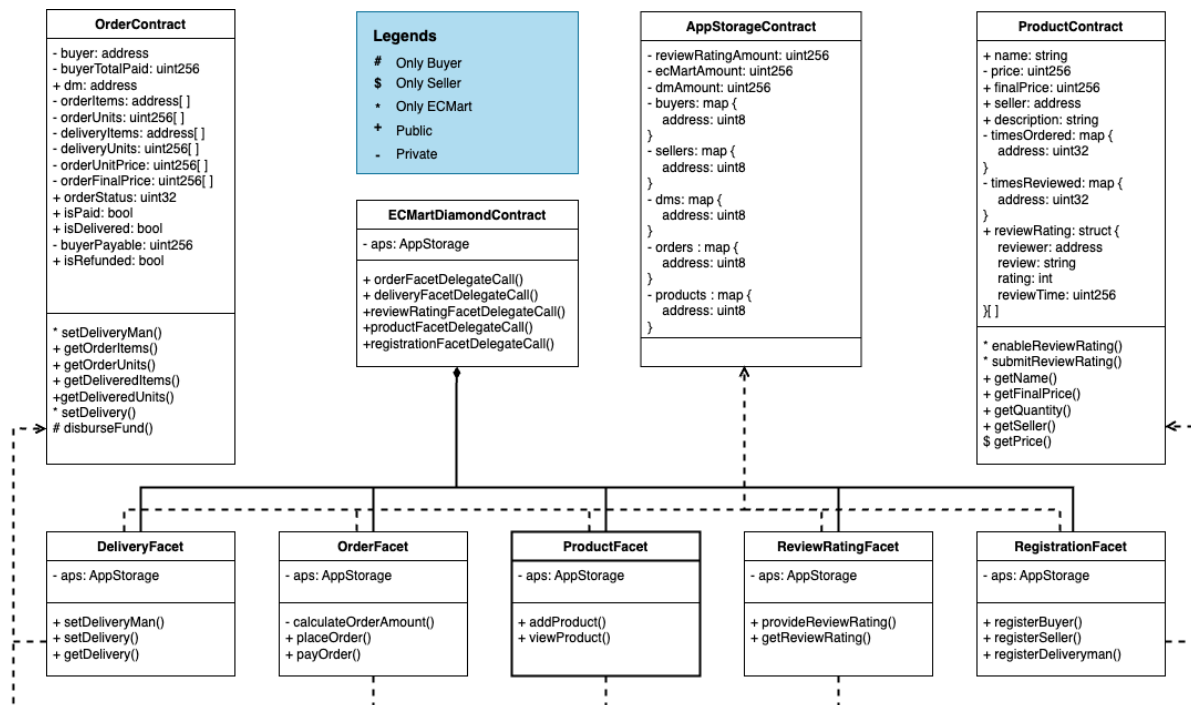


Figure 10: Smart Contract Class Diagram

What is stored on-chain and off-chain?

While financial transactions associated with funds are stored on-chain contracts with our native ERC-20 token 'BDTCC', user credentials, and e-commerce media are stored off-chain using encryption. Apart from that, every data of orders and products is stored both on-chain and off-chain for two-step data persistence.

How will the blockchain architecture handle compliance or regulation of the application if it is for a regulated industry such as banking or insurance?

The architecture of smart-contracts consists of incentivising its stakeholder, and generating revenue for the platform founders. The necessary governmental compliance regarding VAT, TAX is being maintained during operations and sent to the platform account which will later be sent to the government's BTCCC account at the end of each financial year.

How is data stored?

E-commerce heavy media is stored in the IPFS distributed file system. One copy of the e-commerce credential is stored in BigchainDB with encryption, which is a decentralised database. Another copy of data is stored efficiently in the smart-contracts.

Is there a digital identity system in place?

Yes, Decentralised identity is registered for every stakeholder during registration via the Metamask wallet address. And validation of the identity is ensured through the designed

smart-contract at the initiation of every operation or transaction in the e-commerce. Hence, it establishes a firm foundation of both enhanced security and digital identity.

Governance

Our decentralised marketplace aims to achieve a balance between user participation and efficient platform management. Here's an outline of the proposed governance structure:

- **Network Membership Governance**
 - **Open Participation:** The marketplace will be open to anyone with a crypto wallet compatible with the SKALE network and ERC-20 token.
 - **Identity Verification:** Know-Your-Customer (KYC) and Anti-Money Laundering (AML) checks will be implemented for user verification to mitigate fraud and ensure responsible platform usage.
 - **Reputation System:** A user reputation system based on past transactions and reviews will be established. This incentivizes positive behaviour and allows the community to identify trustworthy users.
- **Technology Infrastructure Governance**
 - **Core Developer Team:** A core developer team will be responsible for implementing approved changes and ensuring the smooth operation of the platform.
- **Business Network Governance**
 - **Dispute Resolution:** A decentralised dispute resolution system will be implemented. Users can submit disputes to a community-elected arbitration committee for resolution. This committee will leverage on-chain data and community reputation scores to make fair judgments.
 - **Marketplace Rules Enforcement:** Smart contracts will be designed to enforce core marketplace rules automatically. For instance, contracts can prevent fraudulent listings or ensure timely delivery confirmations.
 - **Community Feedback:** The marketplace will maintain open communication channels to gather user feedback and suggestions for improvement.

Asset Tokenization

The proposed marketplace will introduce its own cryptocurrency, named "BDTCC" (e.g., BanglaCoin). This token will be an ERC-20 token built on the SKALE Network, ensuring compatibility with existing Ethereum wallets and exchanges. Crucially, "BDTCC" will be pegged to the Bangladeshi Taka (BDT) through a basket of reserve assets.

- **Price Stability:** By pegging the token to BDT, product prices will remain stable and familiar to Bangladeshi users, eliminating confusion and simplifying the user experience.
 - We assume 1 BDTCC 'coin' - 1 Bangladeshi Taka.
- **CryptoCurrency Ban Tackle:** As cryptocurrency is banned by Bangladesh Bank from 24 December 2017, In order to unleash the blockchain potential and provide a feasible alternative our native ERC-20 token can facilitate the transactions of the ecommerce. Since, The token circulation rights will be regulated by Bangladesh Bank

the token can be considered as a safe and stable alternative by concurrently maintaining the law.

Privacy and Security Risks

Privacy

1. Is data privacy and identity privacy addressed by the solution?

Yes, Data on off-chain is stored in a decentralised database BigchainDB in encrypted format. Therefore, the data is kept secure and persistent due to decentralisation as well as privacy is ensured via encryption even after being compromised. At the blockchain end, the buyer/seller of the orders can only be viewed by stakeholders who are associated with Order, which is verified by the transaction wallet address of the user.

2. Is there a risk of unintentional data leakage to unauthorised parties within the solution?

No, data leaks to unauthorised parties has been prevented by checking the transaction initiator wallet address in every marketplace action done in blockchain. As the SKALE chain has strong security measures and access controls, unintentional data leakage to unauthorised parties is not an option.

Crypto-security

1. How does the solution handle key management, and does it utilise a standardised and reliable key management system?

Users hold private keys securely in Metamask wallets for access and transactions. Ethers.js facilitates interaction without exposing private keys to smart contracts. Moreover, Cryptographic techniques like digital signatures, encryption, and hash functions ensure that transactions and sensitive information are securely stored and transmitted.

Access Control

1. Are there mechanisms in place within the system to control and restrict access to specific parts of the system?

Buyers, sellers, delivery men have specific access levels. Only authorised users can interact with specific functions of the smart contract.

Valuation & Distribution

This diagram outlines a comprehensive framework for a blockchain-based e-commerce platform, emphasising strategic partnerships with regulatory bodies. Key activities include enhancing platform features, conducting targeted marketing and so on. The platform provides increased security, transparency, less fraud, and competitive benefits in the marketplace, all underpinned by a committed staff, well-trained engineers, and sufficient money. Platform development, staff, marketing, compliance, and operating costs are covered by costs, while transaction fees, commissions, subscriptions, advertising, and premium services provide revenue.

The Bangladesh E-commerce Market generated a significant 56,870 crore taka in 2021 [5]. By leveraging blockchain technology, we aim to capture a small fraction of the market (5% of

sellers and buyers with transactions of around 300 taka per buyer annually). Based on that we featured a profit-loss statement of the first year which is projected below.

Business Model Canvas

ECMart: Blockchain-powered Decentralised E-commerce Marketplace



Figure 11: Business Model of Blockchain-based e-commerce

We have forecasted revenue, expenses, and net income/loss for the initial three years. As expected for a startup, expenses are high in year one due to upfront costs. As the project moves into its peak years two and three, these costs should progressively decline, increasing the anticipated net revenue. Using blockchain technology, which promotes a safe and transparent environment, increases revenue by attracting customers and fostering confidence. This trust, in conjunction with effective operations, will drive us towards a quick break-even point and, ultimately, significant wealth generation.

A detailed business valuation of the proposed e-commerce blockchain marketplace is provided in **Appendix B**.

	Financial Year 1	Financial Year 2	Financial Year 3
Income	15,612,500	23309002	29309002
Expense	16,728,000	13562937	16562937
Net Income / (Loss)	-1115500	9746065	12746065

Table 1: Profit Projection of first 3 years of Blockchain-based e-commerce

ECMART			
Income Statement			
For The Financial Year 2024-2025			
Details	Taka	Taka	Taka
Revenue:			
Commission Fee (5%)		10,237,500	
Advertisement Commission		3105000	
Listing Fee		1705000	
Value Added Service Fee		565000	
Total Revenue			15,612,500
Expense:			
Operational Cost		2760000	
Technology and Maintainance		2268000	
Marketing and Advertising		1380000	
Logistics and Fulfillment		2070000	
Initial Investment Expenses (One Time)		8250000	
Total Expense			16,728,000
Net Profit / Loss Year 2024			-1115500

Figure 12: Income Statement of ECMart for the year 2024

We have forecasted revenue, expenses, and net income/loss for the initial three years. As expected for a startup, expenses are high in year one due to upfront costs. As the project moves into its peak years two and three, these costs should progressively decline, increasing the anticipated net revenue. Using blockchain technology, which promotes a safe and transparent environment, increases revenue by attracting customers and fostering confidence. This trust, in conjunction with effective operations, will drive us towards a quick break-even point and, ultimately, significant wealth generation.

Reasons for Launch

In today's e-commerce environment, trust and transparency pose significant challenges. Conventional marketplaces often depend on middlemen for payment processing and dispute resolution, which can result in higher costs and slower transactions. Our blockchain-based marketplace is designed to overcome these issues by utilising decentralised technology to

create a secure, transparent, and efficient platform. Below we highlight the key reasons for launch:



Figure 13: Launch Phases

- To address significant trust issues and fraud in the traditional e-commerce marketplace.
- To provide a secure, transparent, and decentralised platform using blockchain technology.
- To eliminate problems like fake reviews, non-payment, and non-delivery crimes, which have impacted consumer confidence and market growth.

Conclusion

As the landscape of digital commerce continues to grow, our blockchain-based marketplace presents a groundbreaking answer to the age-old problems of trust and transparency. By tapping into the power of blockchain technology and smart contracts, we're building a secure, efficient, and cost-effective platform that's designed with users in mind. Without the need for middlemen, we're able to cut transaction costs and speed up the process, making it simpler and faster for people to do business. The smart contracts add an extra layer of security by ensuring sellers are only paid once buyers have received their goods, which greatly reduces the chance of fraud. This not only makes the buying and selling process more straightforward but also creates a fairer, more trustworthy marketplace. Looking ahead, we believe this innovative platform has the potential to reshape the way online commerce is done, setting a new standard for trust and efficiency in digital transactions and empowering users to engage in a marketplace where they can transact with complete confidence.

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Appendix A

A github repository is attached below consisting of the off-chain codebase, database, smart-contracts, and contract testing codes mentioned.

<https://github.com/osmanhassan/ECMart>

Appendix B

An excel spreadsheet is attached which includes an entire business valuation with necessary calculation.

[Business Valuation_Chain Reaction](#)